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Article Title: Planned Load Reduction *Versus* Fixed Load: A Strategy to Reduce the Perception of Effort With Similar Improvements in Hypertrophy and Strength

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Planned load reduction *versus* fixed load: a strategy to reduce the perception of effort with similar improvements in hypertrophy and strength

Submission type: Original Investigation.

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Running head: Load reduction training on hypertrophy and strength.

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Abstract

Purpose: The purpose of this study was to compare muscle thickness and 10RM between no load reduction and load reductions during 16 weeks of resistance training (RT). **Methods:** Twenty-one moderately trained men (age: 23.2±4.2 years, body mass: 75.1±7.6 Kg, height: 175±4 cm) were randomized into one of three exercise groups: Control (CON, n=7), all sets with 10RM load; 5% load reduction (RED5, n=7), and 10% load reduction (RED10, n=7) for set 2 and set 3. The RT program consisted of completed three sets each of Biceps and Scott Curls, performed to volitional fatigue $3d \cdot wk^{-1}$. **Results:** Volume load lifted over the 16wk was similar among groups (CON: 38,495±4,397 kg; RED5: 37,388±3,684 kg; RED10: 42,634±6,733 kg; p=0.094). Muscle thickness increased in all groups (p<0.001), with no differences noted among groups (p=0.976). Biceps Curl and Scott Curl 10RM increased in all groups (p<0.001), with no differences noted among groups (Scott Curl: p=0.238; Bicep Curl: p=0.401). RPE was significantly lower for RED 10 (6.8±0.1) as compared to CON (7.0±0.1; p<0.001) or RED5 (7.1±0.1; p=0.001) for the Scott Curl. RPE was significantly lower (p=0.001) for the Biceps Curl in RED 10 (6.8±0.3) than in CON (7.3±0.9), with neither group different from RED 5 (7.0±0.1). **Conclusion:** load reduction did not yield a difference in hypertrophy or 10RM as compared to CON. However, RED10 induced a significantly lower RPE. Thus, load reduction may be a beneficial strategy to reduce the perception of effort during training while achieving similar improvements in hypertrophy and strength.

Keywords: Resistance exercise; training; workload; muscle hypertrophy, muscle strength.

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Introduction

Skeletal muscle mass is a critical biomarker for maintaining health and optimizing athletic performance. Increasing skeletal muscle mass is a major goal of many exercise regimens. There is a debate about the most effective training model to optimize muscle hypertrophy. Manipulation of several training variables (intensity level, numbers of sets, number of repetitions per set, velocity of execution, rest intervals between sets and sessions, exercise order, etc.) has been proposed as necessary to optimize gains in muscle hypertrophy [1,2].

The number of repetitions per set, for example, has been popularly proposed as a vital program component for ensuring optimum hypertrophy. For example, the ACSM points to support from evidence category A (Evidence from well-designed randomized control trials that provide a consistent pattern of findings in the population for which the recommendation is made) [2,3] that the optimal number of repetitions to optimize hypertrophy should be between 8 and 12 repetitions per set for each exercise. This proposal is shared by several other authors [4–7]. However, it has anecdotally been proposed that a certain degree of fatigue must be accrued during each work set, independent of the number of repetitions performed. Thus, some have proposed that we should not establish a fixed number of repetitions, rather, trainees should work to failure (voluntary exhaustion) during every work set [8]. It is still controversial which method would be more efficient to optimize muscle hypertrophy. The major difficulty has been to develop a methodology that the two proposals can be efficiently compared [9,10].

When training to failure, the relationship between volume and intensity must be preserved. Thus, when choosing the number of repetitions to perform, load must be equivalent to allow complete execution of the movements of each set. This format is difficult to fulfill, as has been amply demonstrated in the literature that a given set's performance is affected by previously performed sets. Fatigue from previous sets performed results in a reduction in the

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number of repetitions in subsequent sets. The length of rest interval between sets crucially affects the performance, with the obvious inverse relationship between rest interval and amount of work that may be performed [11].

Load reduction, in which the amount of weight lifted is reduced in successive sets, is one strategy to preserve total volume that an individual may complete in the presence of accumulating fatigue. It is unknown whether the additional volume allowable via load reduction would offset the decrease in intensity in providing the optimal stimulus for producing hypertrophy. Furthermore, in our knowledge, no studies had studied planned load reduction on muscle skeletal adaptations, volume workload and perceived exertion.

The purpose of this study was to compare muscle thickness and 10RM between no load reduction and load reductions during 16 weeks of resistance training. We hypothesized that load reduction would result in a greater volume load lifted, and produce greater increases in muscle thickness than a control condition.

Methods

Subjects

Twenty-one men (age: 23.2±4.2 years, body mass: 75.1±7.6 Kg, height: 175±4 cm) with at least two years of recreational resistance training experience participated in the current study (Table 1). All subjects were characterized by the following training history; consistent participation in a resistance training program during the previous two years with a minimum training frequency of three sessions per week; one hour per session; three to five sets per exercise; six to fifteen repetitions per set; Biceps curl and Scott experience; repetitions maximum sets (to failure) experience; and one to two minutes rest intervals between sets.

Additional exclusion criteria were: a) subjects could not be using drugs or nutritional supplements that could affect repetitions performance for at least three months prior to the

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experiment; b) subjects could not exhibit bone, joint or muscular problems that could limit the effective execution of biceps exercise; c) subjects could not be performing any extraneous structured exercise activity for the duration of the study. All participants read and signed an informed consent, which thoroughly explained the testing procedures, and the experimental procedures were approved by local ethics committee.

Experimental Design

The current study was conducted over 17 weeks; during the first week, anthropometric measures (e.g. height, body mass) were collected and 10RM biceps curl and biceps Scott tests were repeated 72 hours apart to verify reliable loads. During each of the succeeding 16 weeks, subjects performed 48 resistance exercise sessions that involved performance of both biceps exercises. Subjects were separated in 3 groups: a) a constant 10RM load for all three sets of both exercises (CON), b) a 5% reduction following each set (RED 5) (i.e. 10RM, 95% of 10RM, 90% of 10RM, and 85% of 10RM), c) a 10% reduction following each set (RED 10).

The 1-min rest interval between sets has been previously recommended as a training strategy to maximize hypertrophy. The premise behind this being shorter rest intervals increase metabolic and hormonal response to exercise, which may have a positive impact on stimulating greater muscle protein synthesis [12]. Recent work has shown that muscle protein synthesis is likely not impacted by acute changes in circulating hormones after exercise [13]. Nonetheless, multiple sets to failure with short rest intervals are often practiced in hypertrophy training.

Methodology

Each subject completed three exercise session per week for 16 weeks. Each exercise session was conducted on a consistent day and time each week. A strength and conditioning specialist supervised each exercise session to ensure proper technique and provide spotting and verbal encouragement.

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Week 1 was the preparatory period, during which 10RM loads were established for the Biceps curl and Scott curl according to previously published procedures [7]. The 10RM for each exercise was assessed 2 times with 72 hours between tests. Before the 10RM tests, each subject completed 5 minutes of low-intensity aerobic activity (i.e., jogging/walking). Two warm-up sets preceded testing of each exercise at 50% of the perceived 10RM load for 10 repetitions each. After the warm-ups sets were completed, the load was increased to the perceived 10RM, and 1 set was performed to voluntary exhaustion (i.e., muscle failure). The same spotters closely supervised each 10RM attempt, and subjects were instructed to give a verbal signal when voluntary exhaustion was reached. If less than or more than 10 repetitions were accomplished during a given 10RM attempt, the load was adjusted during the next testing session.

The 10RM loads established during the preparatory period were used to design the subsequent 2 weeks testing sessions. Weeks 2, 4, 6, 8, 10, 12 and 14 all subjects performed 10 RM test and, when necessary, loads were corrected. Three times a week (Monday, Wednesday and Friday) subjects completed training sessions with the following load conditions: (a) constant load for all sets (CON), (b) 5% load reduction after each set (RED 5), and (c) 10% load reduction after each set (RED 10). Subjects were assigned to each condition in a randomized fashion.

Each training session during the data collection period began with 5 minutes of low-intensity aerobic activity (i.e., jogging/walking). Two warm-up sets of biceps curl were performed at 50% of the predetermined 10 RM for 15 repetitions each.

Three minutes after the warm-up sets, 3 consecutive sets were performed to the point of voluntary exhaustion (i.e., full repetition maximums). Subjects were allowed exactly 1 minute of rest between sets and 3 min between exercises. The rest intervals were precisely controlled through the use of a handheld stopwatch.

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Ultrasound imaging was used to obtain measurements of muscle thickness (MT). A trained technician performed all testing using a B-mode ultrasound imaging unit (LOGIQ S8, GE, USA). Following a generous application of a water-soluble transmission gel to the measurement site, a linear array probe (Model ML6-15; 12 MHz) was placed perpendicular to the tissue interface without depressing the skin. Equipment settings were optimized for image quality and held constant between testing sessions. When the quality of the image was deemed to be satisfactory, the technician saved the image to hard drive and obtained muscle thickness dimensions by measuring the distance from the subcutaneous adipose tissue-muscle interface to the muscle-bone interface. Measurements were taken on the right side of elbow flexors (combination of biceps brachii and brachialis) upper arm. Measurements were taken 60% distal between the lateral epicondyle of the humerus and the acromion process of the scapula. For each measurement, the examined limb was secured to minimize unwanted movement (Figure 1). In an effort to help ensure that swelling in the muscles from training did not obscure results, images were obtained 48-72 hours before commencement of the study and after the final training session. To further ensure accuracy of measurements, at least 2 images were obtained for each site. If measurements were within 10% of one another the figures were averaged to obtain a final value. If measurements were more than 10% of one another, a third image was obtained and the closest two measurements were then averaged.

Statistical analysis

All data are presented as means ± standard deviation (SD). Data were tested for normality of distribution using Kolmogorov-Smirnov tests. Repeated Measures Analyses of Variance (2 time points x 3 groups) were used to compare differences in 10RM and Biceps thickness over the 16 week period. Total volume load (reps lifted x kg load) completed during the training sessions for both exercises was computed and compared among groups using a

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One-Way ANOVA. Average daily RPE was calculated over each condition, and compared between groups using a One-Way ANOVA. In the presence of significant main effects, post-hoc analyses were accomplished using Tukey's HSD. Pearson product-moment correlations were used to test the relationships among volume load lifted, rating of perceived exertion, 10RM performances, and changes in biceps thickness. All statistical analyses were accomplished using SPSS version 21 (IBM). Statistical significance was set at $p \leq 0.05$.

Results

Muscle thickness

Biceps thickness significantly increased over the training period (time effect $p<0.001$) in all groups. No between-group differences in the change in biceps thickness were revealed (group x time interaction $p=0.976$). Figure 2 displays the changes in biceps thickness among groups.

Biceps strength

Significant increases in 10RM occurred in all groups (Figure 3), for both the Scott Curl (time effect $p=<0.001$, Figure 3A) and the Biceps Curl (time effect $p<0.001$, Figure 3B). No between group differences were noted (Scott Curl group x time interaction $p=0.238$; Bicep Curl group x time interaction $p=0.401$).

Volume load and RPE

Volume load for the Biceps Curl was similar ($p=0.649$, Figure 4A) among groups (CON: 40,767±3,111 kg; RED 5: 42,926±2,624 kg; RED 10: 41,196±6,733 kg). Likewise, volume load for the Scott Curl ($p=0.094$, Figure 4B) was not different among groups (CON: 38,495±4,397 kg; RED 5: 37,388±3,684 kg; RED 10: 42,634±6,733 kg).

Average daily RPE was significantly different for Biceps Curl ($p=0.002$, Figure 4C) and both Scott Curl ($p<0.001$, Figure 4D) among groups. Tukey's tests revealed a significantly

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lower RPE for the Biceps Curl in RED 10 (RPE: 6.8±0.3) than in CON (RPE: 7.3±0.9), with neither group different from RED 5 (RPE: 7.0±0.1) (Figure 4C). For the Scott Curl, RPE was significantly lower (p=0.001) for the RED 10 (RPE: 6.8±0.1) as compared to CON (RPE: 7.0±0.1; p<0.001) or RED 5 (RPE: 7.1±0.1; p=0.001) (Figure 4D).

Correlations among volume, strength, RPE, and biceps thickness.

Pearson product-moment correlations revealed no relationships (p>0.216) between total volume load lifted, RPE, or change in biceps thickness. Modest correlations were found between volume load lifted for Biceps Curls and change in Biceps Curl 10RM (r = -0.522, p=0.015, Figure 5A) and between RPE for Biceps Curls and changes in 10RM Scott Curl (r = -0.469, p=0.032, Figure 5B).

Discussion

Load reduction has been proposed as a training strategy to increase hypertrophy by enabling trainees to complete a greater number of repetitions than could be achieved by lifting a constant load. In the present study, neither the 5% nor 10% load reductions implemented resulted in a difference in total volume load lifted over the training period. In addition to a similar volume load lifted, load reductions (RED 5 and RED 10) yielded similar improvements in strength and muscle size to CON. Subjects in RED 10 experienced a significantly lower perception of effort, despite the completion of every set to volitional fatigue.

It was expected that load reduction would result in a greater volume load lifted as compared to CON. This hypothesis was not supported. Neither RED10 nor RED 5 produced a significant change in volume load lifted vs. CON, thus it is unsurprising that increases in muscle thickness were similar among the three conditions. Prior studies have related greater volume load lifted with greater anabolic intracellular signaling (phosphorylation of p70S6K and rpS6) for muscle hypertrophy [14–16]. Since volume lifted has been shown to affect the

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anabolic intracellular signaling, it would be expected that a greater volume would produce greater hypertrophy than single set training. However, several training studies (10-14 weeks duration) of resistance training have found no difference between multiple and single sets in provoking skeletal muscle hypertrophy [17–20].

Further, Mitchell *et al*. [20] found no relationship between the extent of intracellular anabolic signaling activated during exercise and hypertrophy. In contrast, others have found multiple sets (greater total workout volume) produce more hypertrophy than a single set [21]. Terzis *et al*. [15] compared the acute effect of one, three and five sets of resistance exercise. The authors found no significant difference in phosphorylation of p70S6K and rpS6 between one and three sets; however, the exercise-derived activation of these kinases occurred only after the execution of three and five sets. Further, five sets were more anabolic than one and three sets. In the current study, only three sets of each exercise were performed. Thus, it is possible that total volume performed among all groups in the current study was insufficient to detect differences in hypertrophy among conditions.

Average daily RPE was significantly lower RED10 vs CON or RED 5 for the Scott Curl. Furthermore, average daily RPE was significantly lower for RED10 vs. CON for the Biceps Curl. This indicates that a 10% load reduction can reduce the perception of effort, despite the completion of each set to volitional fatigue. Further, RED 10 produced equal improvements in both hypertrophy and strength to the other conditions. The modest, negative correlations found between RPE and 10RM strength indicate that a greater perception of effort was mildly related to lower improvements in 10RM. Thus, there was no measurable advantage from experiencing a higher RPE in this study, and RED10 was able to produce similar results with a lower perception of effort. However, we should be reiterated that it is unknown whether the benefits of RED10 will apply to multi-joint exercises (such as squats), and therefore, more research should be conducted to determine if this effect occurs in multi-joint exercises.

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Practical applications

In the present study, load reduction did not alter the total volume lifted during 16 weeks of resistance training. While no significant differences in 10RM nor muscle thickness were noted among groups, RED 10 achieved similar results to the other two conditions while inducing a lower RPE. Achieving equal gains in performance while experiencing a lower perception of difficulty is a practically meaningful for any athlete and these data support the use of RED 10 as a training strategy to promote hypertrophy while reducing effort, which in turn may enhance exercise adherence.

Conclusions

Load reduction did not yield a difference in hypertrophy or 10RM as compared to CON. However, RED10 induced a significantly lower RPE. Thus, load reduction may be a beneficial strategy to reduce the perception of effort during training while achieving similar improvements in hypertrophy and strength.

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Conflict of interest statement

No conflicts of interest, financial or otherwise, are declared by the authors.

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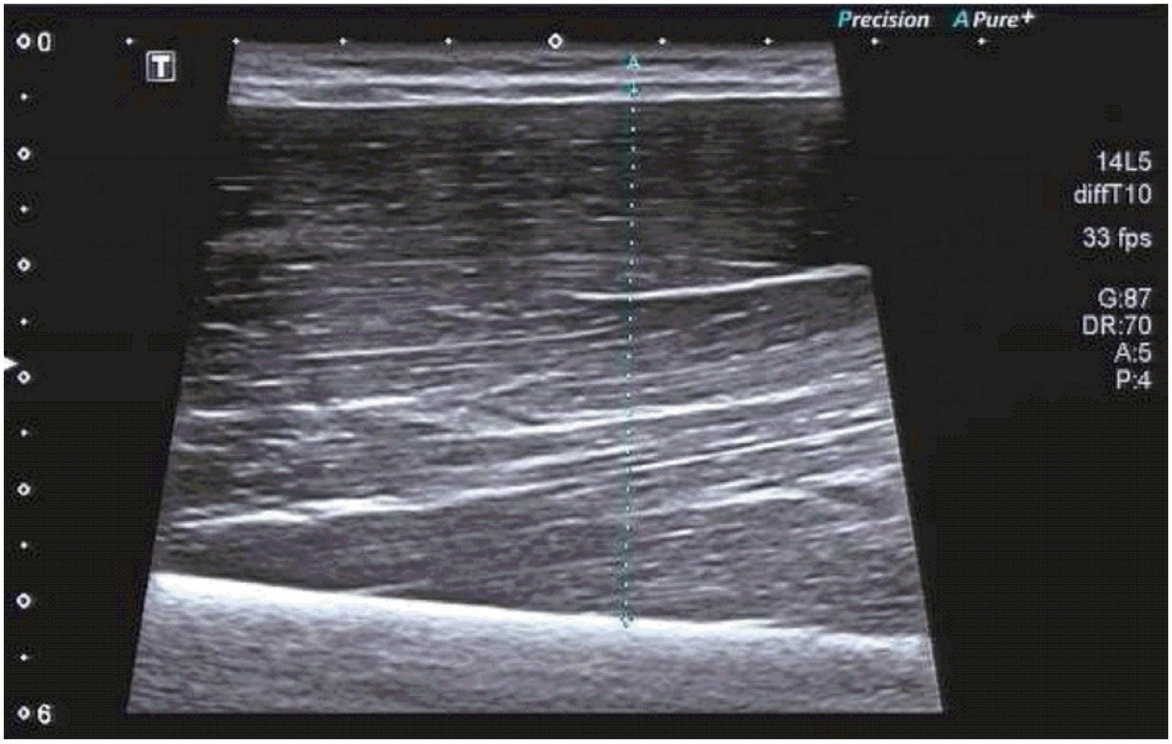

Figure 1. Ultrasound imaging (LOGIQ S8, GE, USA) used to obtain measurements of muscle thickness (MT). A linear array probe (Model ML6-15; 12 MHz) was placed perpendicular to the tissue interface without depressing the skin.

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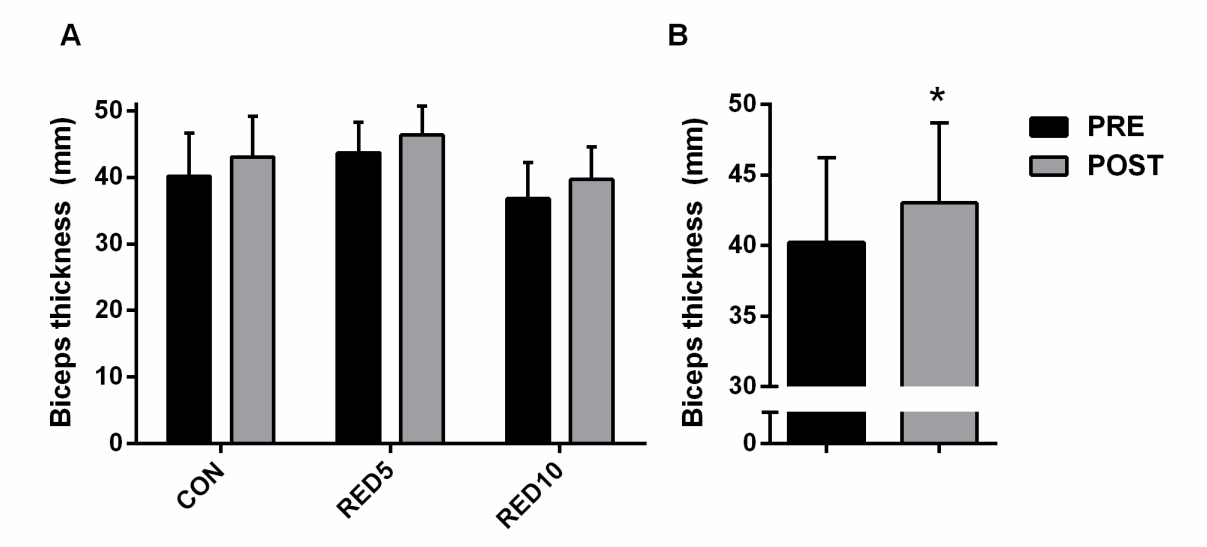

Figure 2. Muscle thickness. Mean±SD for each group (A) and all groups (B). CON, control group; RED5, 5% load reduction group; and RED10, 10% load reduction group. * significant difference ($p<0.05$) to both conditions vs. Pre-training moment.

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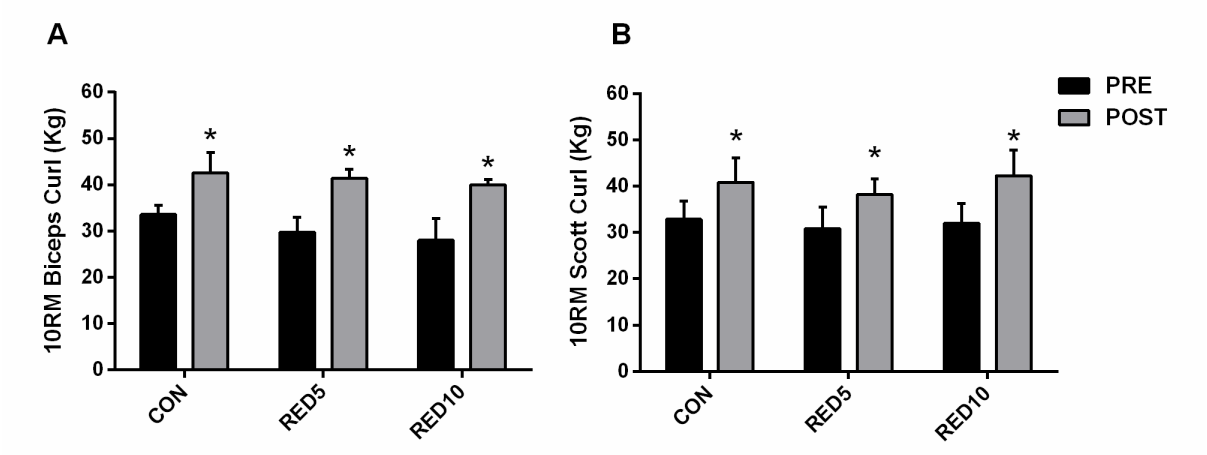

Figure 3. Biceps strength. Mean±SD for each group for 10RM Biceps Curl (A) Scott Curl (B) and all groups. CON, control group; RED5, 5% load reduction group; and RED10, 10% load reduction group. * significant difference ($p<0.05$) to both conditions vs. Pre-training moment.

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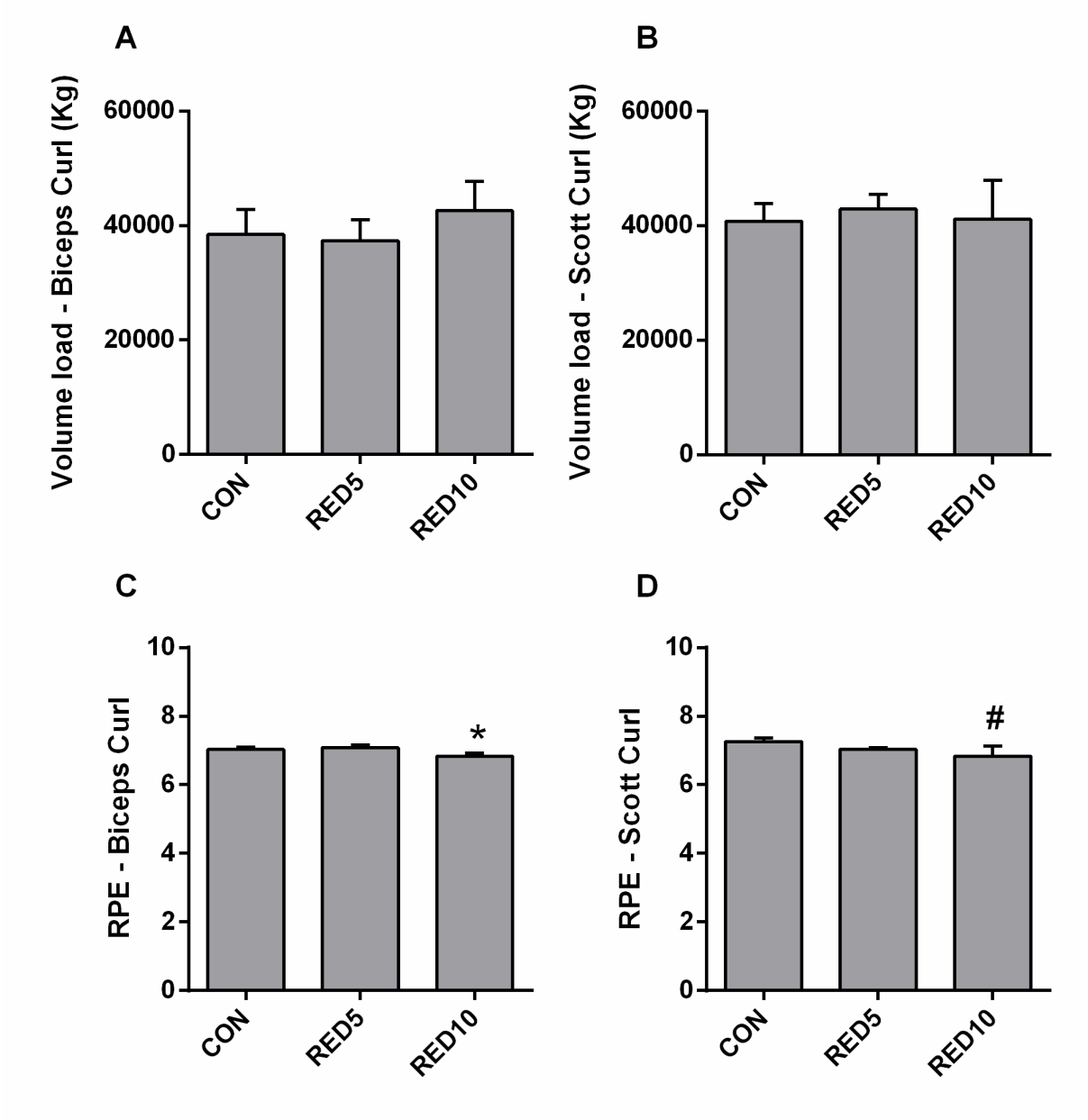

Figure 4. Volume load and RPE. Mean±SD for each group for volume load (A and B) and RPE (C and D) for biceps curl (A and C) and Scott Curl (B and D). CON, control group; RED5, 5% load reduction group; and RED10, 10% load reduction group. * significant difference ($p<0.05$) to RE10 vs. CON, # significant difference ($p<0.05$) to RE10 vs. CON and RED5.

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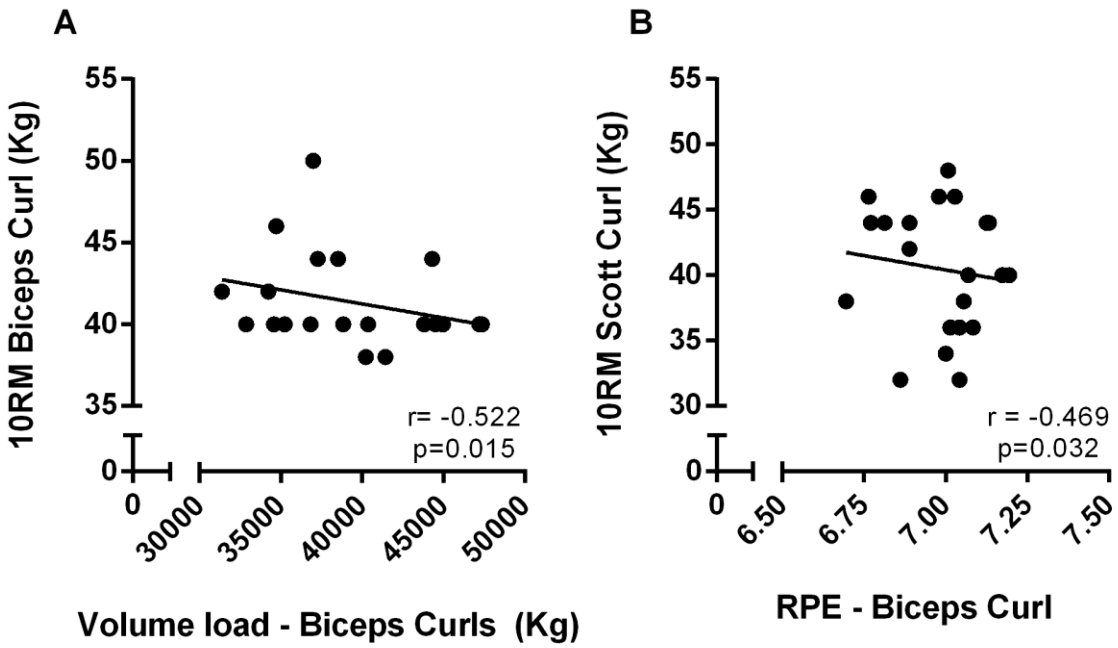

Figure 5. Correlations among volume, strength, and RPE. Pearson correlation between Volume load - Biceps Curls (kg) vs. 10RM biceps curl (kg) (A), and between RPE - biceps curl vs. 10RM Scott curl (kg). The "r" values shown a significant and moderate correlation between variables.

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Table 1. Subject characteristics (n=21).

Variables	CON (n=7)	RED5 (n=7)	RED10 (n=7)
Age (y)	25.3±5.4	22.0±2.1	20.9±3.2
Height (m)	1.76±0.6	1.78±0.7	1.73±0.5
Body Mass (kg)	80.3±3.3	73.4±5.5	73.4±6.7
10RM Biceps Curl (kg)	33.6±2.0	29.7±3.3	28.0±4.8
10RM Scott Curl (kg)	32.9±4.0	30.9±4.6	32.0±4.3

CON, control group; RED5, 5% load reduction group; and RED10, 10% load reduction group.

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